AMARNATH S. PATEL

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EDUCATION

University of Central Florida **Undergraduate Student 4.00 GPA**

B.S. in Physics (Optics & Lasers) and Mathematics, Computer Science Minor (43 credit hours) August 2025 – Present

Florida Atlantic University **3.66 GPA**

University coursework completed via FAU High School, ages 14–18 (108 credit hours; 3 more at Palm Beach State College) August 2021 – May 2025

RESEARCH EXPERIENCE

UCF Astrophotonics Lab, CREOL — Undergraduate Researcher August 2025 – Present

Supervisor: Dr. Stephen Eikenberry

- Measured the wavelength-dependent complex transfer matrix of a photonic lantern by off-axis digital holography, extending Dobias et al., *Opt. Express* **34**(9), 17217 (2026). Aligned the interferometric bench and acquired holograms of the multimode output for each input port across the C-band (1525–1575 nm), recovering full amplitude and phase — one row of the transfer matrix per measurement.
- Implemented the phase-retrieval and mode-decomposition analysis: FFT sideband isolation and demodulation, Butterworth low-pass, joint numerical optimization of mode-field diameter, defocus quadratic phase and field position, then decomposition onto the LP basis to recover complex modal amplitude and phase.
- Produced the per-port reconstruction-fidelity characterization for 6- and 7-port lanterns, reaching ≈ 98% fidelity $|\langle E_{\mathrm{rec}}, E_{\mathrm{sim}}\rangle|^2$ against simulated fields.
- Automated the four-instrument acquisition chain — tunable laser, InGaAs camera, fiber switch, and motorized polarization control — so a complete all-port × C-band sweep runs unattended, with polarization optimized in-loop for peak fringe contrast and saturated frames rejected, reducing a full dataset from days of manual bench time to a single hands-off run.
- Own the telescope-node software for PolyOculus, an astronomical instrument program building photometric observing capability from a networked array of 8 telescopes, each driven by a Jetson Nano.
- Brought the nodes up on mainline Linux 7.1 and Arch Linux ARM, replacing NVIDIA's end-of-life Ubuntu 18.04 image: traced boot failures to an SD-card regulator behind a modular Tegra I2C driver, U-Boot's 14 MiB initramfs load window, and missing Tegra PWM/host1x drivers, and built a kernel that boots with no initramfs. Any node rebuilds from the repository in about 15 minutes (Ansible).
- Wrote centroid-based autoguiding from the DMK guide camera to INDI mount guide pulses, cutting simulated tracking error from 3.01″ to 0.29″ RMS, and closed-loop piezo fast-steering-mirror control that holds a laser spot to 0.2 px.

UCF Physics Department — Undergraduate Research Assistant March 2026 – Present

Physics Education Research — Dr. Zhongzhou Chen (NSF Award 2421299; Gates Foundation INV-076932)

- Research on automated assessment generation for introductory physics instruction: built ESTELA, a system producing multi-version isomorphic exams from structured problem banks, and an agentic generation–evaluation pipeline that iteratively drafts and validates new problems against instructor criteria.
- Materials authored for the project are in use by external adopters following a faculty workshop. Manuscript in preparation.

FAU Grant-Funded AI Safety Research Project (High School) January 2024 – March 2025

Supervisor: Tucker Hindle, Florida Atlantic University

- Benchmarked 10+ coherence and detection methods (GPT-2 perplexity, RoBERTa, BERT NSP, LSA, NLI, burstiness) across 3,125+ generated sequences; identified GPT-2 perplexity as the strongest discriminator (3.3× separation between coherent and incoherent text).
- Characterized a consistent quality–evasion tradeoff across four adversarial humanization pipelines. Grant-funded with HPC access; presented at the Wilkes Honors College Symposium (2025).

OTHER EXPERIENCE

Secretary — National Society of Black Physicists (NSBP), UCF Chapter 2025 – Present

- Coordinate meetings, records, and communications for the UCF student chapter of the National Society of Black Physicists.

President — Coding Club, Florida Atlantic University (High School) 2023 – 2025

- Founded and led a student coding club; organized meetings and collaborative programming projects with peers.

Teaching Assistant — Calculus (High School) August 2024 – May 2025

- Assisted 70 undergraduate students with learning calculus; office hours, exam review, grading. Part-time (10 h/ week).

Advanced Experimental Vehicles — Programmer, Leader, Builder (High School) November 2023 – May 2025

- Configured Arch Linux ARM on Raspberry Pi 5 with Hyprland compositor and WireGuard VPN, enabling worldwide real-time telemetry monitoring of BMS, GPS, and camera feeds.
- Won 2nd Place in Division and Lockheed Martin Award for "Highest Level of Engineering Excellence."

PRESENTATIONS

Batch Processing for Automated Grading via Azure OpenAI June 2026

University of Central Florida, Downtown Campus. ESTELA project (Physics Education Research, Dr. Zhongzhou Chen).

Coherence and Detection Approaches for Identifying AI-Generated Text March 2025

Wilkes Honors College Undergraduate Research Symposium, Florida Atlantic University.

TECHNICAL REPORTS

Coherence and Detection Approaches for Identifying AI-Generated Text 2025

- Empirical study of AI-text detection and adversarial evasion: benchmarked 10 coherence and detection methods (GPT-2 perplexity best, 3.35× separation; RoBERTa the only effective detector) and built four evasion pipelines, characterizing a consistent quality–evasion tradeoff. Grant-funded, Florida Atlantic University.

SELECTED PROJECTS

CELERIS — Metalens Design Pipeline & From-Scratch RCWA Solver 2026

- From-scratch solver for Maxwell's equations in periodic subwavelength structures via rigorous coupled-wave analysis / the Fourier modal method: 1D TE/TM multilayer gratings with Li/Liu–Fan inverse-rule factorization, full 2D-vectorial $P \cdot Q$ formulation for biperiodic nanopillar arrays, and stable Redheffer scattering-matrix layer recursion.
- Validated every layer of the stack against closed-form physics, an independent method, or energy conservation: RCWA vs. transfer-matrix method to 1e-6, multilayer S-matrix to 1e-15, 2D⟶1D degenerate reduction to 5e-12, Σ (diffraction efficiency) = 1.000000, and diffraction-limited designed focal spots at FWHM = $\lambda f/D$. Independently cross-validated against the grcwa and Stanford S^4 solvers (≈ 1e-7) and reproduces canonical published metalenses (Khorasaninejad et al., *Science* 2016; Chen et al., *Nat. Nanotechnol.* 2018).
- Implemented the physical design and analysis capability standalone RCWA kernels lack: Sellmeier and tabulated n, k dispersion, meta-atom library sweeps with automatic etch-depth selection for full-2π phase coverage, gradient-based (Adam) inverse design, Pancharatnam–Berry geometric-phase and polarization-multiplexed lenses exploiting form birefringence (reporting per-polarization RMS phase and focal isolation in dB), plus Strehl, Zernike wavefront, MTF, chromatic focal shift, through-focus caustic, and fabrication-tolerance Monte-Carlo analysis with fabrication-ready GDSII export.
- Accelerated Rayleigh–Sommerfeld far-field propagation as a CUDA kernel, benchmarked against the project's own optimized 16-core CPU path: 4.8–5.8× at 92k pillars, agreeing to machine precision, with automatic CPU fallback.

Diagnosed the kernel as memory-bound from the speedup's decline with grid size. C++23/CUDA, with Python bindings and a native desktop GUI. *Sole-author manuscript in preparation (Computer Physics Communications).*

Stardust — Archival Recovery of Near-Earth Asteroids October 2026

- Built the orbit and detection pipeline: from only the sightings that existed 4 days after discovery, a batched GPU N-body integrator in PyTorch (DE440 perturbers, solar relativity; 100,000 clones × 1 year in 23 s) samples the orbit uncertainty and projects it onto years of pre-discovery ZTF images, searched by a 1.9M-parameter U-Net trained on 4,000 real cutouts and a classic streak filter.
- Blind backtest on 72 asteroids once on NASA's Sentry impact-risk list: 167 verified, never-connected sightings of 22, tightening their orbits up to 52× (median 7.1×); a frozen held-out round recovered 16 of 50. The U-Net finds 57.5% of asteroids at the camera's limit, where the streak filter finds none. Validated against JPL on 2024 YR4 (2032 Earth miss 279,443 km versus JPL's 278,333 km).

TECHNICAL SKILLS

Instrumentation: GPIB/VISA, RS-232, INDI, GigE Vision camera acquisition, tunable IR laser control, motorized polarization control, interferometric alignment, FITS & GDSII data formats

Scientific Computing: FFT & signal processing, phase retrieval & modal decomposition, electromagnetic simulation (RCWA), GPU/CUDA acceleration, NumPy, SciPy

Programming Languages: C/C++, CUDA, Python, Shell (Fish, Bash, tcsh), LaTeX/Typst

Tools: Linux (incl. embedded kernel bring-up on NVIDIA Jetson), Git, CMake, Ansible, Qt/PySide6

SELECTED COURSEWORK

Physics: Electricity & Magnetism I (in progress), Mathematical Methods for Physics, Modern Physics, Quantum Information Processing, Geometric Optics & Lab

Mathematics: Linear Algebra (proof-based, in progress), Partial Differential Equations (in progress), Complex Variables, Matrix & Linear Algebra, Statistical Methods

Computer Science: Deep Learning, Data Structures & Algorithm Analysis, Computer Architecture, Discrete Structures

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HONORS & AWARDS

President's Honor Roll, University of Central Florida (Fall 2025, Spring 2026) 2025 – 2026

Florida Bright Futures — Florida Academic Scholars (highest tier; 100% tuition) 2025

Lockheed Martin Award — "Highest Level of Engineering Excellence," AEV Competition 2024

2nd Place in Division, Advanced Experimental Vehicles Competition 2024

1st Place, Night Hacks Hackathon — UniUtils (course schedule generator & classroom finder) 2023